Shared face estimates and triangular gluing for planar link diagrams

Abstract

The exact polynomial of a marked planar link diagram uses one variable for each face, shared by all its appearances. We give finite criteria on these common variables that bound every noncentral part of the invariant over the full real spectral cube. A weak, non-strict balance condition at the complete circle packing suffices: a common power change of the face weights gives a linear entropy gain and only a quadratic cost. A second theorem preserves arbitrary unabridged sector certificates under an oriented triangular gluing, retaining the single coefficient target at the joining face. Both criteria imply the volume limit for independently sufficiently long fixed signed fillings, using numbered finite, analytic and geometric companion theorems. An explicit seventeen-region certificate and its iterated tetrahedral attachments give marked diagrams outside weak balance. The proof includes the finite-cover logic and the rigorous interval verification of this example.

1 Common face weights and the volume statement

A local estimate for a crossing box becomes useful only after its variables have been identified with those of its neighbors. Choosing a separate modulus for each occurrence of a face would generally estimate a different coefficient. We keep this identification throughout, both when changing the complete packing weights and when joining two diagrams.

Our input D is a finite oriented simple triangulation of the sphere, with at least four vertices, together with a perfect matching of its cubic dual. The R matched pairs comprise $2R$ triangles and $R+2$ vertices. At pair e write the cyclic quadrilateral as (x_e, b_e, y_e, c_e), with matching diagonal $x_e y_e$. Its polynomial arguments are ordered (x_e, b_e, c_e, y_e). Choose a nonmatching edge uv. The two corresponding face variables are *protected*: $L_u = L_v = 0$. Let V be the set of the R remaining, ordinary faces.

The inverse-cut construction of [5, Section 2] gives an augmented link in S^3 with these exact labels. Each matched pair records a crossing circle, its twice-punctured disk and its ordered two-strand collar. Retain $\epsilon_e \in \{0, 1, -1\}$ half twists in that collar. Filling its primitive slope $\mu_e + p_e\lambda_e$ inserts the crossing power

$$n_e = \epsilon_e - 2p_e. \tag{1.1}$$

The local parity pairings and the outside connections are part of the input. In every knot statement below, following them must give one main component, with no additional closed main component.

Write Y_D for the complete augmented exterior. Its circle packing has zero curvatures at u, v and positive curvatures w_f^* at ordinary faces. The scale of all ordinary curvatures is immaterial. We use the absolutely normalized triangle potential

$$\begin{aligned} \Lambda(t) &= -\int_0^t \log|2\sin s|\, ds, \\ e(a) &= \frac{\Lambda(\pi a) + \Lambda(\pi/2 - \pi a)}{\pi}, \qquad 0 \le a \le \tfrac{1}{2}. \end{aligned} \tag{1.2}$$

$$\mathcal{Q}(w_1,w_2,w_3)=\max_{\substack{a_i\ge 0,\ \sum_i a_i=1/2\\ a_i=0 \text{ if } w_i=0}}\left\{\sum_i e(a_i)+\sum_{w_i>0} a_i\log w_i\right\}. \tag{1.3}$$

At least one weight in each triangle is positive. In particular, $\mathcal{Q}(0,0,B)=\frac{1}{2}\log B$. No logarithm of a protected weight is taken. At a positive ordinary vector L, put

$$\begin{aligned} X_e=A_{e,\mathrm{C}}&=\mathcal{Q}(x_e,b_e,y_e)+\mathcal{Q}(x_e,c_e,y_e),\\ A_{e,\mathrm{L}}&=\max\{\log c_e,\mathcal{Q}(x_e,b_e,y_e)+\tfrac{1}{2}\log c_e\},\quad c_e>0,\\ A_{e,\mathrm{R}}&=\max\{\log b_e,\mathcal{Q}(x_e,c_e,y_e)+\tfrac{1}{2}\log b_e\},\quad b_e>0. \end{aligned} \tag{1.4}$$

Here a face label in a function denotes its weight in L. The three labels describe actual disjoint path sectors, defined in Section 2. Left is absent if c_e is protected; right is absent if b_e is protected. Call the remaining set S_e and set

$$\Sigma_D=\left(\prod_{e=1}^{R} S_e\right)\setminus\{\mathrm{C}^R\},\qquad \Phi_D=\frac{\mathrm{Vol}(Y_D)}{2\pi}.$$

Thus Σ_D consists of all locally allowed noncentral strings. It includes strings whose eventual shared coefficient happens to be zero.

For a nonempty set $B\subseteq\Sigma_D$ define the common-weight cost

$$G_{D,B}(L)=\max_{\omega\in B}\sum_e A_{e,\omega_e}(L)-\sum_{f\in V}\log L_f. \tag{1.5}$$

For a single string we write $G_{D,\omega}$. Every occurrence of the same face in this expression uses the same coordinate L_f.

Theorem 1.1 (Finite shared-face criterion)**.** *Let D be as above. Suppose a finite collection of nonempty sets $B\subseteq\Sigma_D$ covers Σ_D, and each set has a fixed positive ordinary vector L^B, such that*

$$g=\min_B\{\Phi_D-G_{D,B}(L^B)\}>0. \tag{1.6}$$

Then there is a threshold T_D, depending on these fixed data, such that every fixed integer vector $\boldsymbol{p}$ with $\min_e|n_e|>T_D$, and with the stated knot pairing, gives a hyperbolic knot $K_D(\boldsymbol{p})$ satisfying

$$\lim_{N\to\infty}\frac{2\pi}{N}\log|\langle K_D(\boldsymbol{p})\rangle_N|=\mathrm{Vol}(S^3\setminus K_D(\boldsymbol{p})). \tag{1.7}$$

The invariant is the reduced zero-framed Kashaev invariant, with unknot value one. The limit is over all integer levels, including both parities and composite levels. The signs and relative sizes of the fixed long powers are unrestricted. If Σ_D is empty, no witnesses are required and the same conclusion holds.

The theorem concerns the unchanged recorded diagram. It imposes no tree or separator condition, but it does require a strict common-weight certificate. We prove two ways of obtaining one. Weak balance at the complete packing permits equality in every local comparison (Theorem 4.3). Protected triangular gluing preserves any certificate of the stated kind (Theorem 5.3). The explicit seed in Theorem 6.1 shows why the latter hypothesis is more useful than restricting every input to weak balance.

Belted sums of augmented link complements have a substantial geometric theory; see Morgan et al. [8]. Van der Veen [9] proved the volume conjecture for Whitehead chains. We do not identify our marked operation with all such constructions, or claim that its unmarked knot types avoid other known presentations. The assertion here is closure of shared source estimates with the original targets and primitive fillings retained. No classification result from that literature is used in the proof.

2 The exact source and the imported foundations

This section specifies the finite object being bounded and the numbered companion results used later. In particular, the central asymptotic is not an assumption about the whole invariant.

For $N \geq 3$, set $d = N - 1$, $\theta = \pi/N$, and

$$s_j = 2\sin(j\theta), \qquad \chi_N(z) = 2T_N(z/2) + 2, \qquad T_N(\cos t) = \cos Nt.$$

On $1, z, \ldots, z^d$, define the banded polynomial operator

$$\mathsf{C} z^j = 2\cos((2j+1)\theta) z^j - s_j^2 (x+y) z^{j-1} - s_j s_{j-1} xy\, z^{j-2}.$$

Negative powers vanish. The finite Fourier theorem [1, Proposition 4.1] identifies this operator, after the specified specializations, transpose and gauge, with each physical residue block of the two-strand Casimir. Its characteristic polynomial is χ_N. Use the polynomial divided difference

$$V_N(\sigma; \mathsf{C}) = \frac{\chi_N(\sigma) - \chi_N(\mathsf{C})}{\sigma - \mathsf{C}}, \qquad H_V(x, b, c, y; \sigma) = \sum_{0 \leq i \leq j \leq d} V_N(\sigma; \mathsf{C})_{ij} b^i c^{d-j}.$$

Division here is polynomial division, including at repeated roots. The face polynomial has total degree d, and spectral degree at most d.

Expanding a triangular inverse before clearing its denominator gives decreasing paths $j \to i$ with steps of length one or two. The monomial $b^i c^{d-j}$ accompanies the outer step weights and every unused spectral row. For $h = \lfloor d/2 \rfloor$, partition them by

sector	condition
C	$i \leq h, \quad d - j \leq h$
L	$d - j > h$
R	$i > h.$

(2.1)

Since $i \leq j$, two one-sided conditions cannot hold together. This partition is exhaustive. At odd levels its central zero-step path appears once, and a path skipping the central row keeps its unused spectral factor. Denote the resulting polynomials by $H_{V,s}$. A protected overloaded middle kills the corresponding one-sided polynomial exactly.

Take the single original coefficient

$$S_{N,\omega}(\boldsymbol{\sigma}) = \left[\prod_{f \in V} L_f^d\right] \prod_{e=1}^{R} H_{V,\omega_e}(L_{x_e}, L_{b_e}, L_{c_e}, L_{y_e}; \sigma_e)\Big|_{L_u = L_v = 0}. \tag{2.2}$$

Write $S_N = \sum_{\omega \in \prod S_e} S_{N,\omega}$, $C_N = S_{N,\mathrm{C}^R}$, and $D_N = S_N - C_N$. Each has degree at most d in each separate spectral variable. Only the fixed number of factors is used here; the total spectral degree need not be at most d.

Proposition 2.1 (Finite evaluation and its norm)**.** *For the original marked diagram and any integer powers n_e, its raw reduced ribbon contraction is*

$$Z_N(\boldsymbol{n}) = \left(\bigotimes_e \mathcal{L}_N^{n_e}\right) S_N, \qquad Z_{N,\mathrm{C}} = \left(\bigotimes_e \mathcal{L}_N^{n_e}\right) C_N. \tag{2.3}$$

For every complex polynomial P of separate degrees at most d,

$$\left|\left(\bigotimes_e \mathcal{L}_N^{n_e}\right) P\right| \leq \prod_e (1 + |n_e|/2)\, \|P\|_{[-2,2]^R}. \tag{2.4}$$

The prescribed zero-framing and enhanced-component corrections have modulus one. Thus they do not change the exponential rate of Z_N.

Imported proof and its hypotheses. The coefficient identity is [1, Theorem 9.1 and Corollary 9.2]; the invariant comparison is its Theorem 10.1. These statements retain the actual cups and caps, unrestricted reduced mark, primitive face lattice, and even-level affine coset. In particular there is no extra factor N^R multiplying (2.2). Theorem 1.1 of that paper gives the one-variable norm; successive application gives (2.4). Its functional uses both values and first normals at repeated spectral roots, and the value at the odd simple root. The separate-degree condition just verified is exactly its hypothesis. Negative powers are included. □

Proposition 2.2 (Uniform local envelopes)**.** *For any fixed positive ordinary vector L, with the protected coordinates zero, and each locally allowed label s,*

$$\limsup_{N\to\infty} \frac{1}{N} \log \sup_{\substack{|Z_f|=L_f \\ \sigma\in[-2,2] \\ (f\in V),\ Z_u=Z_v=0}} |H_{V,s}(Z_x, Z_b, Z_c, Z_y; \sigma)| \le A_{e,s}(L).$$

The supremum includes every face phase and every real spectrum, and the limit superior includes both level parities.

Imported proof and its hypotheses. This is [2, Theorem 1.1]. The required $x_e + y_e > 0$ holds: in a simple sphere the unique edge between u, v is the nonmatching marked edge, and so it cannot also be the matching diagonal. The theorem retains both unused products for the one-sided sectors and the odd skipped-row term for the central sector. No complete packing condition is required of L. □

Proposition 2.3 (Nonzero central contribution and its volume)**.** *The complete weights satisfy*

$$P(q) := \sum_e X_e(e^q) - \sum_{f\in V} q_f, \qquad P(q^*) = \min P = \Phi_D, \quad \nabla P(q^*) = 0, \quad q_f^* = \log w_f^*. \tag{2.5}$$

For independently sufficiently long fixed signed powers the original central contraction is eventually nonzero and has logarithmic rate

$$\lim_{N\to\infty} \frac{1}{N} \log |Z_{N,\mathrm{C}}| = \operatorname{Re} S_{\boldsymbol{n}}, \qquad 2\pi \operatorname{Re} S_{\boldsymbol{n}} = \operatorname{Vol}(S^3 \setminus K_D(\boldsymbol{p})), \qquad \operatorname{Re} S_{\boldsymbol{n}} \longrightarrow \Phi_D \tag{2.6}$$

as all $|n_e| \to \infty$, independently.

Imported proof and its hypotheses. The complete packing, positive labeled cut and stationary action (2.5) are supplied by [5, Theorem 1.1, Proposition 2.1 and Section 3]. Convexity of $\mathcal{Q}$ makes that stationary point a global minimum. The ordinary graph is connected, and the ordinary difference rows generate the primitive common-scale quotient. Each matched pair has a positive outer weight and at least one nonmonomial triangle. These are the geometric hypotheses of [3, Theorem 1.1]; it proves the nonzero holomorphic shared coefficient expansion and the full complementary spectral strip bound for C_N.

With its notation the expansion has unit $\varepsilon_N = (-1)^{dR}$, level power $\alpha = (3R+1)/2$, and coefficient $(2\pi)^{-(R-1)/2}/\sqrt{\det H_N}$. Theorem 1.1 of [4] evaluates the entire original filling sum under exactly those two source hypotheses. Section 7 of [3] records the normalization: its second parity factor is $(-1)^{(N+1)R}$, so their product is one. The actual leading term is

$$\Gamma_N (-2\mathrm{i})^R \sqrt{2\pi}\, N^{R+1/2} \frac{\prod_e \sin(2\pi z_{N,e})}{\sqrt{\det H_N(z_N)}\sqrt{\det Q_N}}\, e^{N S_{N,\boldsymbol{n}}}, \tag{2.7}$$

with relative error tending to zero, nonzero limiting coefficient after these level and unit factors, and $S_{N,\boldsymbol{n}} \to S_{\boldsymbol{n}}$. Here z_N is the unit-charge saddle, H_N the face Hessian, Q_N the

negative spectral Hessian, and $\Gamma_N = \exp(\pi \mathrm{i} d^2 \sum_e n_e/(4N))$. The finite action stays inside the exponential.

Finally [5, Theorem 1.1 and Corollary 8.1] identifies this limiting action with the smooth filling of the original disk slopes. The local potentials agree by their complete values and their continued spectral derivatives, as calculated in [2, Lemma 5.1]. The geometric proof fixes the small logarithms before multiplying by the large slopes, proves $U_e + p_e V_e = -2\pi\mathrm{i}$, keeps the main cusp complete, and compares volume on the same filled fundamental chain. These numbered results give (2.6); a formal critical point alone would not. □

3 Finite certificates and the whole remainder

Definition 3.1 (Unabridged certificate)**.** An unabridged certificate for D, with gap $g > 0$, assigns a positive shared ordinary vector L^ω to every $\omega \in \Sigma_D$, such that

$$G_{D,\omega}(L^\omega) \le \Phi_D - g.$$

Only the local protected-overloaded labels have been removed from Σ_D. In particular no string is omitted because of a separate global vanishing or cancellation argument.

A finite cover as in Theorem 1.1 gives this definition by assigning each string to a cover member containing it. Conversely, one witness per string is a finite cover. For a Cartesian branch $B = \prod_e S_{B,e} \subseteq \Sigma_D$, its cost is

$$G_{D,B}(L) = \sum_e \max_{s \in S_{B,e}} A_{e,s}(L) - \sum_{f \in V} \log L_f.$$

For a non-Cartesian branch, the maximum in (1.5) must retain its correlations. Passing to the Cartesian hull is an upper bound, not an equality in general. Choosing a branch of a local maximum is also not the same as selecting an actual path sector.

Lemma 3.2 (Homogeneity and full spectators)**.** *For $\lambda > 0$, every local rate increases by $\log \lambda$ when all its positive weights are multiplied by λ. Consequently every $G_{D,B}$ is unchanged by common rescaling. Moreover, with protected logarithms omitted,*

$$\begin{aligned} \max\{A_{\mathrm{C}}, A_{\mathrm{L}}, A_{\mathrm{R}}\} &= \max\{X, \log b, \log c\} =: H, \\ \max\{A_{\mathrm{C}}, A_{\mathrm{L}}\} &= \max\{X, \log c\} =: H_{\neg \mathrm{R}}. \end{aligned} \tag{3.1}$$

Proof. Every supported simplex in (1.3) has total load $1/2$, so $\mathcal{Q}(\lambda w) = \mathcal{Q}(w) + \frac{1}{2}\log\lambda$. There are R local rates and R ordinary targets, proving the first assertion. Testing the simplex vertex at b gives $\mathcal{Q}(x, b, y) \ge \frac{1}{2}\log b$; similarly for c. Thus every mixed term in (1.4) is at most X. This proves both exact maximum identities. □

Lemma 3.3 (Shared coefficient bound)**.** *A certificate with gap g gives*

$$\limsup_{N\to\infty} \frac{1}{N} \log \|D_N\|_{[-2,2]^R} \le \Phi_D - g. \tag{3.2}$$

Proof. For a string ω, apply the multivariate Cauchy coefficient formula to (2.2) on its one shared torus $|Z_f| = L_f^\omega$. Proposition 2.2 gives

$$|S_{N,\omega}(\boldsymbol{\sigma})| \le \prod_{f \in V} (L_f^\omega)^{-d} \prod_e \sup_{\text{same torus}} |H_{V,\omega_e}(Z; \sigma_e)|.$$

On dividing the logarithm by N, the difference between d/N and 1 contributes only $N^{-1}\sum_f \log L_f^\omega$. Hence the limit superior of the uniform spectral rate is at most $G_{D,\omega}(L^\omega)$.

There are finitely many strings and fixed witnesses, so all local errors are simultaneous for sufficiently large N. Finally $D_N = \sum_{\omega\in\Sigma_D} S_{N,\omega}$; the triangle inequality costs only the fixed factor $|\Sigma_D|$. This proves (3.2) through both parities. No infimum over weights has been interchanged with a maximum over strings. □

Proof of Theorem 1.1. Use (2.4) on D_N. The powers are fixed, so its prefactor has zero exponential rate. Thus

$$\limsup_N N^{-1}\log|Z_N - Z_{N,\mathrm{C}}| \le \Phi_D - g.$$

Increase the independent long-filling threshold until all conclusions of Proposition 2.3 hold and $\operatorname{Re} S_{\boldsymbol{n}} > \Phi_D - g/4$. For these fixed powers, (3.2) implies an eventual upper rate $\Phi_D - 3g/4$, after absorbing the fixed norm constant and the finite local errors. The central limit gives an eventual lower rate $\operatorname{Re} S_{\boldsymbol{n}} - g/8$. Their ratio therefore tends to zero exponentially. In particular $Z_N \neq 0$ for all large N, and $Z_N/Z_{N,\mathrm{C}} = 1 + O_{\boldsymbol{n}}(e^{-cN})$ for some $c > 0$. The unit normalization in Proposition 2.1 and the actual volume identity (2.6) give (1.7). If Σ_D is empty, the remainder is zero and the same conclusion follows directly. □

4 Weak balance and a quantitative defect criterion

The complete packing is a convenient starting point for a common weight vector. It is not necessary that the final Cauchy weights describe another circle packing.

For an original triangle t, let $I_t \subseteq V$ be its ordinary corners, and write

$$Q_t(q) = \mathcal{Q}((e^{q_i})_{i\in I_t}, 0 \text{ at protected corners}).$$

The scalar function e in (1.2) satisfies

$$e(0) = e(1/2) = 0, \quad e'(a) = \log\cot(\pi a), \quad e''(a) = -\frac{2\pi}{\sin(2\pi a)} < 0.$$

These formulas include the absolute normalization: $\Lambda(\pi/2) = 0$, obtained by reflection and the double-angle identity for the integral of $\log\sin$. Thus e is positive in the interior.

Lemma 4.1 (Triangle loads and curvature)**.** *If $|I_t| \ge 2$, set $w_i = e^{q_i}$ and*

$$D_t^2 = \sum_{\{i,j\}\subseteq I_t} w_i w_j, \qquad a_{ti} = \frac{1}{\pi}\arctan(w_i/D_t).$$

These are the unique maximizing loads in (1.3). They obey

$$\partial_i Q_t = a_{ti}, \qquad E_t(q) := Q_t(q) - q\cdot\nabla Q_t(q) = \sum_i e(a_{ti}) > 0.$$

For a singleton row, $Q_t = q_i/2$ and $E_t = 0$. In either case

$$h^{\mathsf{T}}\nabla^2 Q_t(q)h = \sum_{\{i,j\}\subseteq I_t} \kappa_{ij}^t (h_i - h_j)^2, \qquad 0 < \kappa_{ij}^t = \frac{w_i w_j}{2\pi D_t(w_i + w_j)} \le \frac{1}{4\pi}. \tag{4.1}$$

Proof. For two entries the two tangent ratios are reciprocal. For three, $D_t^2 - w_i w_j = w_k(w_i + w_j)$ and the tangent addition formula show that the displayed angles sum to $\pi/2$. Also $e'(a_{ti}) + q_i = \log D_t$. Strict concavity gives the maximizer; its interior location follows from the infinite positive endpoint derivative, so the entropy is strictly positive. Differentiating the angles gives the off-diagonal entry $-\kappa_{ij}^t$; with three entries the cancellation is $D_t^2 + w_i^2 = (w_i + w_j)(w_i + w_k)$. Logarithmic homogeneity gives zero Hessian row sums, proving the quadratic form. Finally $D_t \ge \sqrt{w_i w_j}$ and $w_i + w_j \ge 2\sqrt{w_i w_j}$ give its bound. □

From now on $q = \log w^*$ is the fixed complete vector. Define

$$\mathcal{E} = \frac{1}{4\pi} \sum_t \sum_{\{i,j\} \subseteq I_t} (q_i - q_j)^2 = \frac{1}{2\pi} \sum_{ij \text{ a white edge} \atop i,j \in V} (q_i - q_j)^2. \tag{4.2}$$

The equality counts the two triangles at each ordinary edge. It is invariant under changing the common complete scale. For $0 < t < 1$ use the shared vector $L_f(t) = e^{(1-t)q_f}$. Taylor's formula, (2.5) and (4.1) give

$$0 \le P((1-t)q) - \Phi_D = \int_0^t (t-s) q^{\mathsf{T}} \nabla^2 P((1-s)q) q \, ds \le \tfrac{1}{2} t^2 \mathcal{E}. \tag{4.3}$$

There is a useful direct form of the entropy gain. The function Q_t is convex, being a supremum of affine functions of q. Its supporting tangent at q gives, for $0 < \beta \le 1$,

$$Q_t(\beta q) \ge Q_t(q) + (\beta - 1) q \cdot \nabla Q_t(q) = \beta Q_t(q) + (1-\beta) E_t(q). \tag{4.4}$$

This also holds exactly for singleton rows. No concavity assertion about a maximum of different local sectors is involved.

If there is no allowed one-sided label, then Σ_D is empty and Theorem 1.1 already applies. For the rest of this section suppose at least one such label survives. For each allowed side $s = \mathrm{L}, \mathrm{R}$ at box e, let z be its overloaded middle (c_e for left, b_e for right), and let $t(e,s)$ be the triangle containing z. Put

$$\epsilon_{e,s} = \max\{0, q_z - X_e(e^q)\}, \qquad h_{e,s} = E_{t(e,s)}(q). \tag{4.5}$$

Every $h_{e,s} > 0$. Indeed z is ordinary and positive, while at least one outer coordinate is ordinary because its matched edge cannot be the protected nonmatching edge. Thus this triangle is not a singleton.

Lemma 4.2 (Loss for each side)**.** *For every allowed one-sided label and* $0 < t < 1$*,*

$$(X_e - A_{e,s})(L(t)) \ge t h_{e,s} - (1-t)\epsilon_{e,s}. \tag{4.6}$$

Proof. For left, writing $Q_b = Q(x,b,y)$ and $Q_c = Q(x,c,y)$,

$$X - A_{\mathrm{L}} = \min\{X - \log c, \; Q_c - \tfrac{1}{2}\log c\}.$$

At q, the first entry is at least $-\epsilon_{e,\mathrm{L}}$. The second is nonnegative, by the simplex vertex at c. Apply (4.4) with $\beta = 1 - t$. The first entry gains $t(E_b + E_c)$, at least $t h_{e,\mathrm{L}}$, and the second gains $tE_c = t h_{e,\mathrm{L}}$. Both are bounded below by the right side of (4.6). Right is the same calculation with b, c exchanged. □

Theorem 4.3 (Weak balance and controlled defects)**.** *Suppose for some* $0 < t < 1$

$$\gamma(t) = \min_{(e,s) \text{ allowed}} \{t h_{e,s} - (1-t)\epsilon_{e,s}\} - \tfrac{1}{2} t^2 \mathcal{E} > 0. \tag{4.7}$$

Then $L(t)$ is an unabridged certificate with gap $\gamma(t)$; hence Theorem 1.1 applies. In particular it suffices that the complete packing be weak-balanced*:*

$$X_e(w^*) \ge \log w_b^* \quad (b \in V), \qquad X_e(w^*) \ge \log w_c^* \quad (c \in V) \tag{4.8}$$

at every box, with equality allowed.

Proof. Let $J(\omega)$ be the one-sided positions of a noncentral string. Its cost at $L(t)$ is exactly $P((1-t)q)$ minus the sum of the corresponding local losses. Equations (4.3) and (4.6) give

$$G_{D,\omega}(L(t)) \le \Phi_D + \tfrac{1}{2}t^2\mathcal{E} - \sum_{(e,s)\in J(\omega)} \{th_{e,s} - (1-t)\epsilon_{e,s}\}.$$

Condition (4.7) makes each summand positive, and $J(\omega)$ is nonempty. This proves the certificate. Under (4.8), all defects vanish. Let $h_* = \min h_{e,s} > 0$. If $\mathcal{E} > 0$, choose

$$t_0 = \min\{1/2, h_*/\mathcal{E}\}; \quad \text{if } \mathcal{E} = 0, \quad t_0 = 1/2.$$

Then $\gamma(t_0) \ge t_0 h_*/2 > 0$. If there are no allowed one-sided labels, the certificate is vacuous. □

The quantitative statement also allows positive balance defects: with this t_0, the sufficient condition $\max \epsilon_{e,s} < t_0 h_*/2$ gives a gap at least $t_0 h_*/2 - \max \epsilon_{e,s}$. All constants are computed on the original complete packing. They are not claimed uniform over growing diagrams.

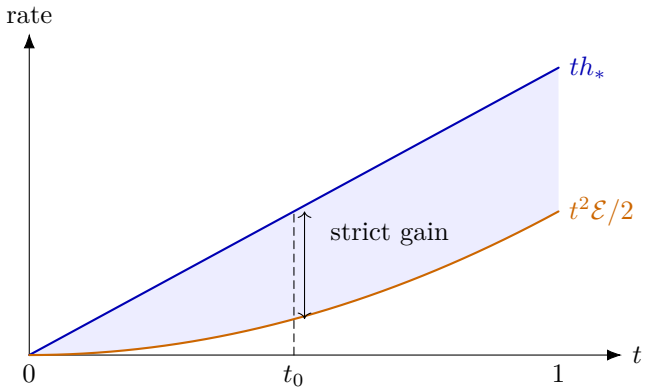


Figure 1: The common-weight change $q \mapsto (1-t)q$. The lower bound for one local loss is linear; the upper bound for the full action cost is quadratic. The drawing uses $h_* = \mathcal{E} = 1$ only to show their orders. In a diagram both constants are defined by its shared ordinary face weights and original triangles.

5 Protected triangular gluing

We now permit different witnesses for different strings. This flexibility makes a local gluing inequality survive the global coefficient extraction.

Take two input spheres. Choose one triangle incident to each marked edge and identify their boundary vertices as u, v, B, where u, v are protected and B is ordinary. Align their matched sides and record the opposite orientations as

	removed oriented face	old polynomial quartet
D_1	(u, v, B)	(u, v, c, B)
D_2	(u, B, v)	(u, a, v, B).

(5.1)

If an entire input must be reflected to arrange this convention, transport its cyclic orders, left/right labels and signed collars with the reflection. The entropy bounds are unchanged. Interior vertices from the two inputs are distinct.

Delete the two faces, identify their boundaries, and pair their old partners (u, a, B) and (u, B, c). The new quartet is

$$(u, a, c, B). \tag{5.2}$$

Every other oriented face and matching pair is retained. The punctured input spheres are disks; joining their boundaries with opposite orientations gives an oriented sphere. There are no new repeated vertices or edges apart from the identified boundary ones, so the triangulation is simple. The new matching edge is uB, and the retained uv is nonmatching. Every triangle is paired once. The counts are

$$R = R_1 + R_2 - 1, \qquad |V_{\text{all}}| = R + 2, \quad |F| = 2R, \quad |E| = 3R. \tag{5.3}$$

The inverse cut therefore realizes an actual new augmented link in S^3, with its own primitive disks. This is an operation on the recorded diagrams; it is not an asserted factorization of their filled invariants.

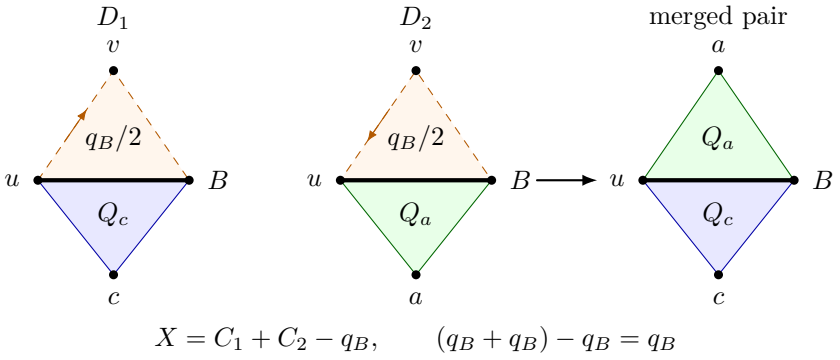


Figure 2: Only the shaded dashed faces are removed. Their boundaries are identified with opposite orientations; u, v remain protected and B becomes one ordinary face with one coefficient target. The thick edge uB now matches the retained triangles. The vertex v, absent from this new pair, remains elsewhere in the sphere. Writing $q_B = \log B$, the removed triangle rates total q_B. This exactly compensates for counting the B-target once rather than twice. The figure records the oriented triangles and their matching.

Lemma 5.1 (The common complete action)**.** *When the two weight vectors agree at B,*

$$P_{\text{new}}(q_1, q_2) = P_1(q_1) + P_2(q_2). \tag{5.4}$$

Consequently $\Phi_{\text{new}} = \Phi_1 + \Phi_2$. The complete weights on each piece are its old weights up to a common rescaling.

Proof. Put $Q_a = \mathcal{Q}(0, a, B)$, $Q_c = \mathcal{Q}(0, c, B)$ and $q_B = \log B$. The two old complete rates and the merged rate are

$$C_1 = \tfrac{1}{2}q_B + Q_c, \qquad C_2 = Q_a + \tfrac{1}{2}q_B, \qquad X = Q_a + Q_c = C_1 + C_2 - q_B.$$

Every other triangle term is unchanged. The two old target sums count q_B twice and the new one once, proving (5.4). Normalize the two complete minima by $B = 1$; they then combine into a global minimum. Conversely the right side is always at least the sum of its two minima. The marked packing's uniqueness up to scale, supplied by [5, Section 3], identifies these complete weights. □

Lemma 5.2 (All three merged sectors)**.** *The new sector has the following locally allowed input strings:*

new label	*input labels*	*comparison*
C	(C, C)	$X = C_1 + C_2 - q_B$
L	(L, C)	$A_{\mathrm{L}} \le L_1 + C_2 - q_B$
R	(C, R)	$A_{\mathrm{R}} \le C_1 + R_2 - q_B$.

(5.5)

Here L_1 and R_2 are the old first-left and second-right rates in their original frames.

Proof. Only left survives at the first old box and only right at the second. Their rates are

$$L_1 = \max\{q_c, \tfrac{1}{2}q_B + \tfrac{1}{2}q_c\}, \qquad R_2 = \max\{q_a, \tfrac{1}{2}q_B + \tfrac{1}{2}q_a\}.$$

The simplex vertices at B give $Q_a, Q_c \ge q_B/2$. Therefore

$$L_1 + C_2 - q_B = \max\{q_c + Q_a - \tfrac{1}{2}q_B, Q_a + \tfrac{1}{2}q_c\} \ge \max\{q_c, Q_a + \tfrac{1}{2}q_c\} = A_{\mathrm{L}}.$$

Exchange a, c for the right inequality. Neither calculation requires q_B to be positive. Both branches of the one-sided maximum have been retained. □

Theorem 5.3 (Closure of unabridged certificates)**.** *If D_1, D_2 have certificates in the sense of Definition 3.1, with gaps $g_1, g_2 > 0$, their protected triangular gluing has one with gap at least* $\min(g_1, g_2)$*. Any fixed finite iteration satisfies*

$$R = \sum_j R_j - (k-1), \qquad \Phi = \sum_j \Phi_j, \qquad g \ge \min_j g_j. \tag{5.6}$$

For every knot-producing parity pattern of the assembled diagram, its independently sufficiently long fixed signed fillings satisfy (1.7)*.*

Proof. Given any noncentral output string, use (5.5) at the merged box and retain every other label on its original piece. This yields locally allowed strings ω_1, ω_2. At least one is noncentral. The exact target cancellation gives

$$G_{\mathrm{new},\omega}(q_1, q_2) \le G_{1,\omega_1}(q_1) + G_{2,\omega_2}(q_2) \quad \text{when } (q_1)_B = (q_2)_B.$$

For a noncentral input choose its certificate vector. For a central input choose its complete minimum. Rescale each whole vector to $B = 1$ using Lemma 3.2. They now agree on all shared variables: the protected zeros and this single ordinary weight. Their costs give the claimed gap, using Lemma 5.1.

This reasoning uses every locally allowed induced string, whether or not that string's old final coefficient vanishes. It would not justify reusing a certificate that discarded such strings. Repeated gluing preserves the simple sphere and the nonmatching mark. At a later step rescale the entire previously assembled vector, not its already identified faces independently. Induction gives (5.6), and Theorem 1.1 supplies the volume conclusion. □

A knot parity is always available, although the theorem also permits any other parity passing the original component test. In the untwisted diagram the physical strands are projection circles. Make a graph with these circles as vertices and a box as an edge joining the two circles meeting its disk; loops are allowed. It is connected because the original four-valent projection is connected. Choose a spanning tree and make exactly its boxes odd. At each step the odd reconnection joins two distinct current circles, so it merges them without producing an extra component. Induction leaves one circle. Either odd sign has this pairing. This does not claim that arbitrary old knot parities remain knots after the gluing.

If both inputs are weak-balanced, the output is also weak-balanced. At complete weights the unmerged comparisons are unchanged, while

$$X = Q_a + Q_c \geq C_1 \geq q_c, \qquad X \geq C_2 \geq q_a.$$

The first inequalities use $Q_a, Q_c \geq q_B/2$. The more general certificate theorem is useful because its inputs need not satisfy weak balance, as the next example shows.

6 A certified seed outside weak balance

We specify the finite example by its actual ordered quartet list, so the argument does not require a diagram compiler to define the source. Let the vertices be $0, \ldots, 18$, the protected edge be $\{0, 16\}$, and take the following polynomial-order quartets:

e	(x, b, c, y)	e	(x, b, c, y)
0	$(1, 0, 4, 2)$	9	$(10, 8, 13, 0)$
1	$(2, 0, 5, 3)$	10	$(11, 9, 14, 12)$
2	$(0, 1, 6, 4)$	11	$(12, 10, 15, 13)$
3	$(4, 2, 7, 5)$	12	$(0, 11, 16, 14)$
4	$(5, 3, 8, 0)$	13	$(14, 12, 17, 15)$
5	$(6, 4, 9, 7)$	14	$(15, 13, 18, 0)$
6	$(7, 5, 10, 8)$	15	$(16, 14, 0, 17)$
7	$(0, 6, 11, 9)$	16	$(17, 15, 0, 18)$
8	$(9, 7, 12, 10)$		

(6.1)

Each row means the oriented faces (x, b, y), (x, y, c), paired along xy. The list gives 34 distinct faces and 51 edges; every edge occurs twice with opposite directions. Each vertex link is one cycle, the face adjacency graph is connected, and $19 - 51 + 34 = 2$. Thus it is an oriented simple sphere. Its 17 matching edges are distinct and pair exactly the listed faces; $\{0, 16\}$ is nonmatching. These finite combinatorial assertions can be checked directly from the table by the procedure in Section 6.3.

Theorem 6.1 (The seed and its attachments)**.** *The marked sphere* (6.1) *has an unabridged forty-branch certificate with gap greater than* $1/25$*. Its complete packing fails weak balance at at least one of the boxes*

$$J = \{2, 4, 7, 12\}.$$

For every fixed $r \geq 0$*, it admits* r *successive tetrahedral attachments that retain those boxes and yield an unabridged certificate on* $17 + r$ *regions, still outside weak balance at the retained mark. All knot-producing long fixed signed fillings of these recorded diagrams satisfy* (1.7)*.*

6.1 The finite cover

The positive rational data used below are the two committed certificates [6, 7]. We describe every kind of entry and the inequalities they certify, including their rigorous evaluation. The saved rational numbers are witnesses; no numerical minimizer is an input to the proof.

There are two kinds of branches. For each nonempty subset of J, require right at exactly those positions of J, allow central or left at the other positions of J, and all locally allowed labels elsewhere. These are 15 branches. If no position of J is right, choose any surviving one-sided label as an anchor, allow central or left at the other positions of J, and all labels elsewhere. There are 25 such anchors. They cover every locally allowed noncentral string: either its right subset in J is nonempty, or it has a one-sided anchor in the second case. No branch contains the all-central string. Overlap causes no problem.

The file [6] stores one positive rational vector for each branch, with the protected zeros retained. Its tags have the following exact meanings:

$$H = \max\{X, q_b, q_c\}, \qquad H_{\neg\mathrm{R}} = \max\{X, q_c\}, \qquad A_{\mathrm{L}}, \quad A_{\mathrm{R}}.$$

These are (3.1), not a replacement of a genuine sector by an invented label. Undefined protected logarithms are omitted.

The same file gives 83 nonnegative rational corner loads α_{ti}, one for each supported triangle corner, with

$$\sum_{i \in I_t} \alpha_{ti} = \tfrac{1}{2}, \qquad \sum_{t \ni f} \alpha_{tf} = 1 \quad (f \in V).$$

Their entropy $E_{\mathrm{low}} = \sum_{t,i} e(\alpha_{ti})$ is a lower bound for Φ_{17}: test these loads in every $\mathcal{Q}$ and cancel the ordinary target logarithms. This is weak duality at every vector, followed by the minimum in (2.5).

The interval calculation detailed below proves

$$E_{\mathrm{low}} > 19.90751095350, \qquad \max_B G_{D,B}(L^B) < 19.85303956990. \tag{6.2}$$

Their difference exceeds $0.05447 > 1/25$. This proves the certificate assertion, including strings that might have zero shared coefficient.

6.2 Locating the failure of weak balance

The second rational file [7] gives a weak-duality lower bound for the smaller comparison

$$\widetilde{G}(q) = A_{12,\mathrm{R}}(q) + \sum_{e \in \{2,4,7\}} H_e(q) + \sum_{e \notin J} X_e(q) - \sum_{f \in V} q_f. \tag{6.3}$$

At spectator $e \in \{2, 4, 7\}$, choose a rational mass $0 < \lambda_e \leq 1$ for its two complete triangle rows and mass $1 - \lambda_e$ on the pure b_e-coordinate. Each weighted triangle row has loads $\alpha_{ti} \geq 0$ summing to $\lambda_e/2$, and entropy $\lambda_e \sum_i e(\alpha_{ti}/\lambda_e)$. At the other non-anchor boxes $\lambda_e = 1$. At anchor 12 the quartet is $(0, 11, 16, 14)$, so

$$A_{12,\mathrm{R}} = \max\{q_{11}, (q_{11} + q_{14})/2\}.$$

Choose loads $1 - \beta, \beta$ on faces $11, 14$, respectively, with rational $0 \leq \beta \leq 1/2$; their entropy is zero. The file contains 84 rational variables in these weighted rows, masses and anchor allocation. Every ordinary face load, including the pure-middle and anchor loads, is exactly one.

The inequalities

$$\max\{X, q_b\} \geq \lambda X + (1 - \lambda) q_b, \qquad A_{12,\mathrm{R}} \geq (1 - \beta) q_{11} + \beta q_{14}$$

and the triangle variational formula show $\widetilde{G}(q) \geq E_{\mathrm{flow}}$ for every q; all logarithmic terms cancel. A separate positive rational vector L^{up} gives $\Phi_{17} \leq P(\log L^{\mathrm{up}})$. Rigorous evaluation gives

$$E_{\mathrm{flow}} > 19.95975965735, \qquad P(\log L^{\mathrm{up}}) < 19.92324700353. \tag{6.4}$$

It follows that $\widetilde{G}(q) > \Phi_{17} + 1/40$ for all q. If all four boxes of J were weak-balanced at the complete vector, then $H_e = X_e$ at $2, 4, 7$, and $A_{12,\mathrm{R}} \leq X_{12}$ by the simplex inequality. This would give $\widetilde{G}(q^*) \leq P(q^*) = \Phi_{17}$, a contradiction.

This comparison does not assert that a fixed genuine sector has rate exceeding volume. The three spectator mixtures are atoms of the upper function H. They serve here only to locate a failure of weak balance in the recorded diagram.

6.3 Reproducible exact and interval verification

The certificate files contain the quartet list, protected and ordinary vertices, rational numbers as integer numerator–denominator pairs, and the row and branch indices just described. The verifier first checks these finite lists and all row and face sums with rational arithmetic. For the topology it counts both oriented occurrences of every edge, traces every vertex link, checks connected face adjacency and Euler characteristic two, and verifies that each matching edge pairs its two declared faces. These are complete finite checks of this particular table, not samples of an infinite family.

For a positive supported triangle, its potential is evaluated by the formula proved in Lemma 4.1:

$$Q_t = \frac{1}{\pi} \sum_{i \in I_t} \{\Lambda(\vartheta_i) + \Lambda(\pi/2 - \vartheta_i) + \vartheta_i \log w_i\}, \quad \vartheta_i = \arctan \frac{w_i}{\sqrt{\sum_{i<j} w_i w_j}}.$$

Singletons are half logarithms. All elementary operations use outward ball arithmetic at 192 bits [10]. For the only infinite series, the absolutely convergent Fourier formula is

$$\Lambda(t) = \sum_{k=1}^{M} \frac{\sin(2kt)}{2k^2} + \mathcal{R}_M(t), \qquad |\mathcal{R}_M(t)| \le \frac{1}{2M}. \tag{6.5}$$

Indeed integrate the Abel-regularized Fourier series for $-\log|2\sin t|$, then pass to the uniformly convergent integrated series. The tail is bounded by $\frac{1}{2}\sum_{k>M} k^{-2} \le 1/(2M)$. Values $e(0) = e(1/2) = 0$ are inserted exactly. Use $M = 2048$ in (6.2) and $M = 8192$ in (6.4). For a maximum of interval enclosures, the lower bound is the maximum of their lower endpoints and the upper bound is the maximum of their upper endpoints. This bounds every branch including ties.

The command

```
python3 papers/planar-certificates/check.py --output /tmp/planar-check
```

replays the two existing exact verifiers, checks the displayed quartet table and orientations, and records all input hashes. It requires Python with `python-flint` and NumPy. The two fixed certificate SHA256 values are

```
54f6087e25e9e6c48e123ea61497cab2a2bb2eb3e4d92a2eb3f9b6e76452eadb
8e9ca9b9baf9383449a54840a4204ac4008b20cd9a56684ffc9a0a45c65bba27.
```

Together with the explicit tail in (6.5), these checks prove the finite inequalities used above. The asymptotic and geometric conclusions are the written implications of the preceding sections, not extrapolations from finite levels.

6.4 Arbitrarily many attachments

The marked-edge triangles of the seed are at boxes 12 and 15. Use the latter's output face:

$$(x, b, c, y)_{15} = (16, 14, 0, 17), \qquad (16, 17, 0).$$

This is the second input of (5.1), with $(u, v, B, a) = (16, 0, 17, 14)$. Thus no box in J is changed.

For the other input take a tetrahedral sphere with oriented faces

$$(u, v, B), \quad (u, B, C), \quad (B, v, C), \quad (v, u, C),$$

matching uB, vC, and protected edge uv. Its quartets are (u, v, C, B) and (v, u, B, C). At $B = C = 1$, each ordinary face receives load $1/2$ from one singleton and $1/4$ from each of the two nonsingleton triangles. It is therefore the complete stationary vector. Each complete box rate is $2e(1/4) > 0$, whereas the positive middle logarithm is zero. It is strictly balanced and has a certificate by Theorem 4.3.

Continue on the newly added side of the retained marked edge, using fresh vertices at each attachment. The original boxes of J remain unchanged. Lemma 5.1 preserves their complete weights up to a common scale, and $X_e - q_b$, $X_e - q_c$ are invariant under that scale. The original failure of weak balance therefore persists. For r attachments,

$$R_r = 17 + r, \qquad \Phi_r = \Phi_{17} + 4r\, e(1/4).$$

Theorem 5.3 provides the unabridged certificate and Theorem 1.1 the knot volume limit. This proves Theorem 6.1. The fixed diagram and its parity pattern are chosen before the independent long fillings and before N.

7 Scope of the criteria

The two sufficient conditions have different uses. Weak balance requires only the original complete packing and supplies one common vector for all strings. Triangular gluing preserves the more flexible assignment of a vector to each genuine string. The seed gives an explicit strict distinction at a fixed mark; it does not exclude another mark or another presentation of the same knot from satisfying weak balance.

The gluing proof does not apply to a bound known only after cancellation of the entire source, nor to a bound with some spectral parameters fixed at their endpoints. Such bounds need not supply the input vectors in (5.5). Likewise, discarding an input string because its coefficient vanishes globally would require a separate support-transfer proof. The unabridged definition avoids that unproved step.

No certificate is asserted for every simple matched sphere. No threshold is uniform over a growing number of pieces or over powers varying with the level. The full volume conclusions proved here use the original spectral functional and primitive fillings of each output diagram. They neither replace the invariant by a positive model nor infer a volume from an unmarked critical point.

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